

RESCURA

Disaster Relief & Rescue Platform

BorNEO HackWknd 2026

Project Proposal & Technical Report

Connecting People. Saving Lives.

A real-time, AI-powered platform designed for the ASEAN context to connect survivors, volunteers, and relief organizations during emergencies.

1. Executive Summary

Rescura is an AI-powered, real-time disaster relief coordination platform built specifically for the ASEAN region. During a disaster, information chaos is as deadly as the disaster itself. Rescura cuts through that chaos by giving survivors a way to shout for help on an interactive map, giving field volunteers turn-by-turn navigation to the most critical incidents, and giving relief organizations a live operations dashboard to deploy supplies and track their teams — all in a single, mobile-friendly web application.

The Problem We Solve

During disasters, survivors have no reliable way to report their exact location and needs. Volunteers waste precious time verifying false reports and finding victims without navigation.

Organizations cannot coordinate supply chains in real time, leading to duplication and shortages.

Families are separated with no way to confirm each other's safety.

Communities lack accessible, localized training to prepare before a disaster strikes.

Rescura addresses every one of these gaps within a single, cohesive platform. It is production-grade, with a real database, real-time messaging, AI triage, and a full role-based access system already implemented.

2. Problem Statement & Context

2.1 The ASEAN Disaster Landscape

Southeast Asia is one of the most disaster-prone regions on the planet. Earthquakes, flash floods, typhoons, and cyclones displace millions of people every year. Myanmar alone has experienced repeated large-scale floods and earthquakes affecting tens of thousands of households annually. The pattern is consistent: when disaster strikes, communication infrastructure breaks down, first responders are overwhelmed, and ordinary citizens have no channel to signal for help at scale.

2.2 The Coordination Gap

Relief efforts consistently suffer from three compounding failures:

Information Vacuum	Survivors cannot communicate precise location or supply needs. Reports travel through social media, phone calls, and word of mouth — unverified, uncoordinated, and often too slow.
Volunteer Inefficiency	Willing volunteers arrive at disaster zones with no system to guide them to the highest-priority victims or track what has already been addressed.
Supply Mismanagement	Aid organizations independently procure and distribute supplies with no shared visibility, creating shortages at some sites and waste at others.
Family Disconnection	Separated family members have no real-time way to check on each other's status, adding mental anguish on top of physical danger.
Preparedness Gap	Most communities lack access to disaster-preparedness training in their native language before a crisis occurs.

3. Our Solution — Rescura

Rescura is a unified platform that turns every smartphone into a node in a disaster-response network. It operates across five interconnected modules:

3.1 Interactive Rescue Map

The map is the operational heart of the platform. Any authenticated user can drop a pin at their GPS coordinates to report a damaged location or declare a safe zone. Every pin carries:

- A photo attachment (stored in Supabase Storage)
- A contact phone number for direct coordination
- A description of the situation and the type of help needed
- AI-suggested supply quantities based on description text and image analysis
- A real-time status badge: Pending → Confirmed → Completed

The map uses Mapbox GL for professional-grade cartography with custom icons, live markers, and popups. Volunteers can request a turn-by-turn driving route from their location to any pin with a single tap, powered by the Mapbox Directions API with haversine fallback.

AI Gate-Keeping

When a civilian tries to submit a Damaged Location pin, Rescura's Gemini AI automatically validates the report.

If the description and photo don't suggest a genuine emergency, the system blocks the submission and asks the user to provide more relevant details.

This dramatically reduces false reports and keeps the map clean for real emergencies.

3.2 Real-Time Alerts & Family Safety

Rescura uses Ably WebSockets to push live alerts to every connected client. Disaster events (earthquakes, floods, cyclones) are broadcast as toast notifications that appear across all active sessions simultaneously. No page refresh is ever needed.

The Family Safety System allows users to register family members by searching for them within the platform. Once linked, a user can:

- See a family member's last-known location on the map
- Send a safety-check notification with a single tap
- Receive real-time confirmations when members mark themselves safe
- Get alerted if a member has not been seen for a configurable window (default: 5 minutes in the field)

3.3 AI-Powered Triage Assistant

Rescura integrates Google Gemini 1.5 Pro as an emergency triage assistant. The AI analyses incident descriptions and attached photos to return:

Severity Score	A 0–1 float indicating urgency (e.g. 0.85 = critical). The UI renders this as a red/yellow/green badge.
Incident Categories	Structural, medical, flooding, fire, general, or critical — enabling rapid sorting.
Supply Recommendations	A ranked list of items and quantities drawn from the platform's live inventory (First Aid, Blankets, Water Bottles, Medicine Box, etc.).
Confidence	The AI's self-assessed certainty, allowing volunteers to weight suggestions accordingly.

The AI layer has a built-in fallback: if the Gemini API is unavailable, a deterministic heuristic engine parses keywords and numbers from the description to generate reasonable supply estimates. This ensures the platform is resilient even in low-connectivity scenarios.

3.4 Volunteer & Organization Management

Rescura has a sophisticated role-based access control system with four distinct user types, each with tailored views and permissions:

Civilian User	Can report damaged locations and safe zones, add family members, browse safety training modules, and receive real-time alerts.
Tracking Volunteer	Field agent deployed by an organization. Can view all pending pins on the map, navigate to victims, confirm reports on arrival (with AI assistance), and log requested supply items against each confirmed pin.
Supply Volunteer	Logistics agent. Sees only confirmed damaged-location pins. Can mark a pin as 'Delivered' once supplies have been distributed, closing the relief loop.
Organization / Admin	Full dashboard with volunteer roster management, supply inventory tracking, cross-organization visibility, and the ability to broadcast disaster alerts to all users via Ably.

Organizations can recruit volunteers directly through the platform, assigning them as 'tracking' or 'supply' roles within their team. The entire volunteer lifecycle — recruitment, deployment, completion — is tracked in the database.

3.5 Safety Education Modules

Rescura includes a gamified safety-education system with modules covering CPR, first aid, flood preparedness, earthquake response, fire evacuation, and more. Each module features:

- Video content with embedded quiz checkpoints
- Q&A reference sections
- Progress tracking with certificate badges on completion
- Difficulty ratings (Beginner / Intermediate / Advanced)
- Module prerequisites to guide a structured learning path

The platform supports both English and Burmese (Myanmar), making it accessible to local communities in the ASEAN region.

4. Technical Architecture

4.1 Technology Stack

Technology	Role	Why We Chose It
Next.js 15	Full-stack Framework	Server-side rendering, API routes, file-based routing, and React 19 support in a single codebase.
Supabase (PostgreSQL)	Database & Auth	Managed PostgreSQL with Row Level Security, real-time subscriptions, and Storage for image uploads.
Mapbox GL JS	Interactive Mapping	Industry-leading vector tile maps with custom markers, popups, and the Directions API for turn-by-turn navigation.
Ably	Real-Time Pub/Sub	WebSocket infrastructure for live alert broadcasting and family-safety notifications across all connected clients.
Google Gemini AI	AI Triage Engine	Multimodal LLM for analysing disaster descriptions + images to recommend supply quantities and severity scores.
Tailwind CSS v4 + shadcn/ui	UI Framework	Utility-first styling with a production-quality component library for consistent, accessible design.
TypeScript	Type Safety	End-to-end type safety across all API routes, data models, and React components.
Deno / Edge Functions	AI Chat Backend	Supabase Edge Functions host the AI chat assistant for low-latency, globally distributed inference.

4.2 Database Schema

The platform is built on 11 relational tables in PostgreSQL:

users	Core identity: name, email, phone, password hash, admin flag.
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organizations	Relief org profiles: name, region, funding, volunteer count, status.
org-member	Many-to-many join: links users to organizations with role (tracking / normal) and status.
pins	Map markers: coordinates, type (damaged/safe), description, phone, image URL, status, confirmed_by.
items	Supply catalogue: name, unit of measure, description.
pin_items	Supplies linked to a pin: requested_qty, remaining_qty — tracks the relief loop.
organization_supplies	Organization inventory: what stock each org holds.
organization_supply_transactions	Audit log of every supply movement.
family_members	Family group links between users.
messages	In-app messaging between users.
user_last_seen	Timestamped location pings for family safety tracking.

Row Level Security (RLS) is configured on all tables, and a public Supabase Storage bucket (pin-images) handles disaster zone photos.

4.3 AI Architecture

The AI triage pipeline is a two-layer system:

- Primary: Google Gemini 1.5 Flash receives the incident description and optional base64-encoded image. It returns structured JSON with severity, categories, and supply items constrained to the platform's live item catalogue.
- Fallback: A deterministic heuristic engine parses the text for critical keywords ('injured', 'collapsed', 'flood', 'fire') and numeric quantities, producing a reasonable supply estimate with no external API call required.

The AI also serves as a gate-keeper: non-rescue reports are blocked before they reach the database, keeping the map free of noise. A separate Supabase Edge Function powers the conversational AI chat assistant, allowing users to ask for first-aid advice, evacuation routes, or emergency protocols in natural language.

5. Key Features at a Glance

Live Rescue Map	Mapbox GL map with color-coded pins (red = damaged, green = safe), status indicators, real-time updates, and a full-screen map experience on mobile.
AI Triage	Gemini-powered severity scoring and supply recommendations on every pin report, with multimodal support for disaster photos.
Turn-by-Turn Navigation	Volunteers can get a driving route from their GPS to any pin. Uses Mapbox Directions API with haversine fallback and an ETA popup.
Real-Time Alerts	Ably WebSocket broadcasts push earthquake, flood, and cyclone alerts as instant toast notifications to all connected users.
Family Safety Check	Register family members, see their last location, and send one-tap safety pings. Alerts fire automatically if members go silent.
Role-Based Dashboards	Four distinct roles (civilian, tracker, supply volunteer, org admin) each with purpose-built views and permissions.
Organization Dashboard	Volunteer roster, supply inventory, org-wide analytics, volunteer creation, and mission tracking in a tabbed admin view.
Safety Education	Gamified CPR, first aid, flood, earthquake, and fire training modules with quizzes, progress tracking, and digital certificates.
Emergency Kit Checklist	Interactive checklist (persisted in localStorage) covering 20 essential emergency preparedness items.
Bilingual Interface	Full English and Burmese (Myanmar) language support across all UI text, navigation, and instructions.
Supply Lifecycle Tracking	Pins track requested vs. remaining quantities of each supply item. Volunteers log deliveries; organizations see live inventory.
AI Chat Assistant	A Gemini-backed conversational assistant accessible globally from all pages — survivors can ask for first-aid instructions, safe routes, or advice at any time.

6. User Journeys

6.1 A Survivor in Need

Step 1	User opens Rescura on their phone. The map centers on their GPS location.
Step 2	Taps 'Add Pin', selects 'Damaged Location', types a description: '3 people injured, building partially collapsed'.
Step 3	Attaches a photo of the damage. Gemini instantly scores severity at 87%, suggests First Aid x3, Blankets x6, Water Bottles x18.
Step 4	Selects their location on the map, enters their phone number, and submits. The pin appears on the map for all volunteers.
Step 5	Receives a message when a tracking volunteer confirms the report on arrival.

6.2 A Field Volunteer

Step 1	Logs in as a Tracking Volunteer. Taps 'Confirm Pin' to see unconfirmed reports within 5 km.
Step 2	Reviews the AI-generated severity scores and picks the most critical case.
Step 3	Taps 'Show Route' to get turn-by-turn navigation to the victim's location.
Step 4	Arrives on site, reviews the victim's photo and description, adjusts the AI-suggested supply quantities as needed.
Step 5	Confirms the pin. The system logs their confirmation and notifies the organization's supply volunteers.

6.3 A Relief Organization

Step 1	Organization admin logs in. Dashboard shows all confirmed pins needing supplies, total volunteers active, and inventory levels.
Step 2	Creates a supply volunteer account and assigns them to confirmed pins with outstanding supply needs.

Step 3	Broadcasts a real-time flood alert to all platform users via the Ably-powered broadcast API.
Step 4	Supply volunteer delivers aid and marks the pin as 'Delivered'. Inventory updates automatically.
Step 5	Organization reviews completed missions and remaining supply gaps in the analytics tab.

7. Impact & Social Value

Rescura directly contributes to United Nations Sustainable Development Goal 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action) by building resilience and reducing the human cost of natural disasters.

Measurable Impact Targets

Response Time Reduction: AI-validated pins reach volunteers faster, cutting average response time for critical cases.

False Report Filtering: AI gate-keeping is estimated to reduce non-emergency pin submissions by 60-80%, keeping the map actionable.

Supply Efficiency: Closed-loop tracking (requested qty vs. remaining qty) eliminates duplicate deliveries and shortages.

Community Preparedness: Safety modules can train thousands of community members before the next disaster strikes.

Family Reunification: Real-time last-seen tracking and safety pings reduce the anxiety of separation.

The platform is architected for ASEAN scale: Supabase's managed PostgreSQL infrastructure handles concurrent users globally, Ably's WebSocket infrastructure is battle-tested for millions of simultaneous connections, and Mapbox's CDN ensures map tiles load fast even on mobile networks.

8. Innovation & Differentiation

Rescura is not a concept or a prototype. Every feature described in this report is implemented in working code. What sets it apart from other disaster apps:

AI-Gated Reporting	Most platforms accept any pin. Rescura uses Gemini to validate reports before they enter the database, maintaining signal quality under pressure.
Multimodal AI Triage	The AI processes both text and images simultaneously — not just keywords — giving more accurate severity scores and supply estimates.
Closed-Loop Supply Tracking	From 'pin created' to 'supplies delivered', every step is logged. Organizations can see exactly what has been done and what is still needed.
Resilient AI Fallback	If Gemini is unavailable (common in disaster zones with poor connectivity), a deterministic heuristic engine takes over seamlessly.
Role-Graduated UX	Rather than showing every user everything, Rescura presents each role with exactly what they need — reducing cognitive load in high-stress situations.
ASEAN-First Design	The platform defaults to Myanmar's geography and supports Burmese language. It is designed for real communities, not a generic global audience.

9. Future Development Roadmap

Immediate (Post-Hackathon)

- Offline-first mode: cache critical map data and queue pin submissions for when connectivity is restored
- SMS integration: allow survivors without data plans to submit pins via text message
- Push notifications: native mobile alerts for high-severity pings near the user's location

Short-Term (3-6 Months)

- Integration with USGS earthquake feeds and national flood-warning APIs for automated alerts
- Supply chain integration with major ASEAN NGOs for real inventory levels
- Multi-language expansion: Thai, Malay, Indonesian, Filipino
- Progressive Web App packaging for offline installation without an app store

Long-Term Vision

- Drone coordination module for aerial survey and supply drops
- AI-powered demand forecasting to pre-position supplies before predicted events
- Government integration: direct data feeds to national disaster management agencies
- Cross-border coordination for multi-country disaster events (typhoons, transboundary floods)

10. Meet the Team

Rescura was built by a dedicated team of five developers who combined their skills in full-stack engineering, UI/UX design, AI integration, and cloud infrastructure to deliver a production-ready platform during BorNEO HackWknd 2026.

#	Name	Role
1	Thwin Khant Oo	Team Lead & Full-Stack Developer
2	Entt Oo Aung	Backend Developer & Database Architect
3	Hlaing Htet Htet Htun	Frontend Developer & UI/UX Designer
4	Aung Sint Naing	AI Integration & Real-Time Systems Engineer
5	Hein Zayar Win	Cloud Infrastructure & DevOps Engineer

Our Mission as a Team

We built Rescura because we believe technology can and should save lives. Every line of code in this platform was written with a real disaster survivor in mind. We are proud to present a working, deployable solution — not just a concept.

11. Why Rescura Should Win

Rescura is not a demo. It is a working, deployable application with real infrastructure, real AI, and real code. Every feature — from the interactive rescue map to the family safety tracker to the AI triage assistant — is implemented and functional.

What Makes Rescura Stand Out — Summary

COMPLETE: Full-stack implementation, not a mockup. Real database, real AI, real real-time infrastructure.

IMPACTFUL: Directly addresses life-or-death coordination failures that cost lives in every major ASEAN disaster.

INNOVATIVE: AI-gated pin reporting and multimodal triage are novel approaches not found in existing disaster apps.

SCALABLE: Built on enterprise-grade infrastructure (Supabase, Ably, Mapbox) that scales from a local emergency to a national disaster.

CONTEXTUAL: Designed specifically for ASEAN communities, with Burmese language support and regional geography as defaults.

DEPLOYABLE: The README includes full setup instructions. Rescura can be live and serving real communities within hours.

Disasters do not wait. Rescura is ready.

Thank you for your consideration.

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